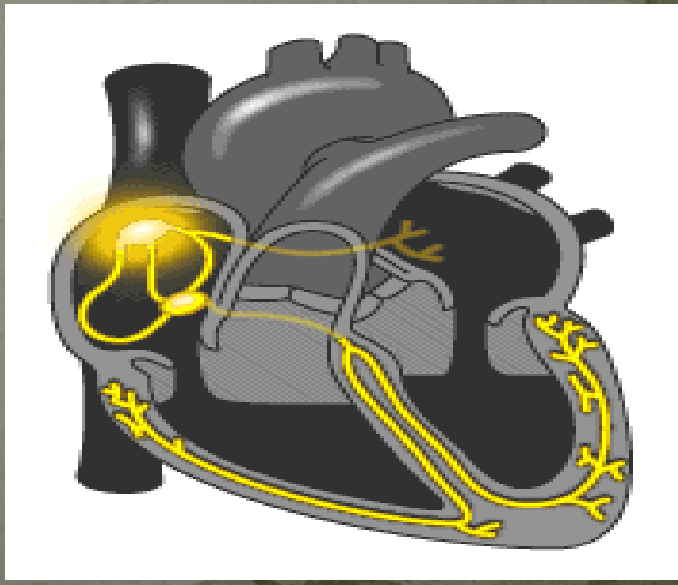
Congenital Heart Diseases (CHD)

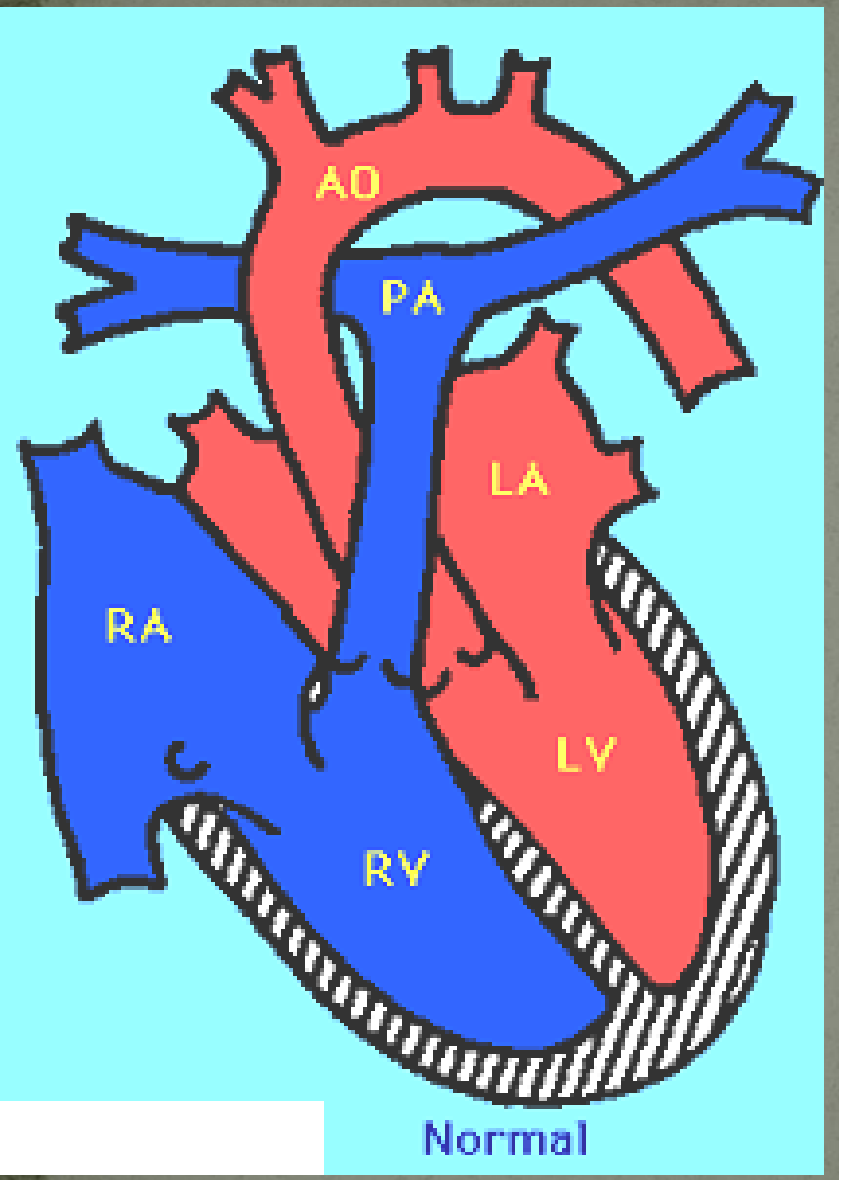
BY

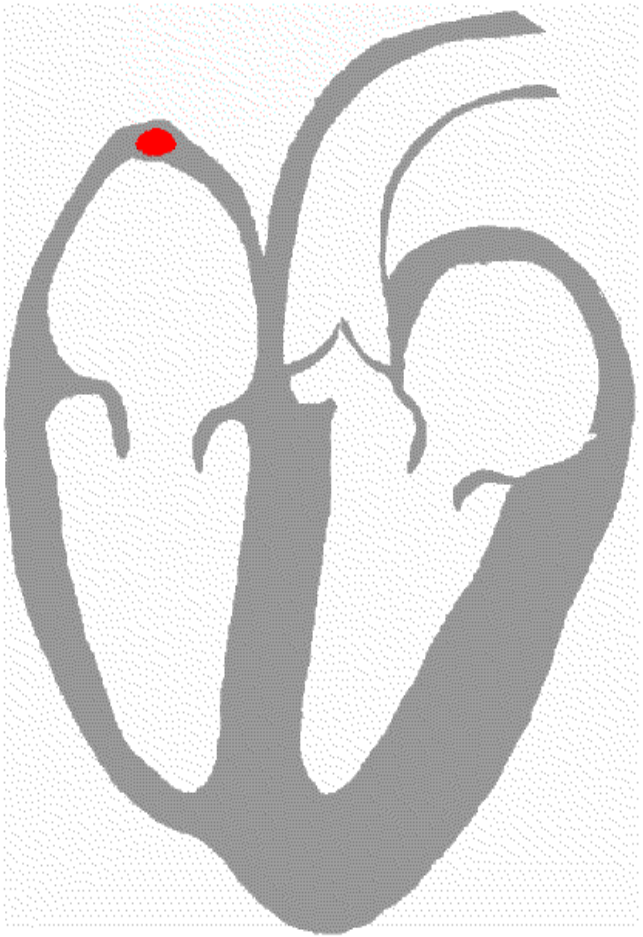
Dr. Safwat M. Abdel-Aziz



Definition:

Congenital heart disease refers to structural or functional heart disease that is present at birth


AO
PA
LA
RA
LV
RV
Normal



–

Classifications of congenital heart diseases

I-Congenital heart disease with permanent cyanosis (i.e. dominant right to left shunt).

1- **Cyanotic heart disease with increased pulmonary flow:**

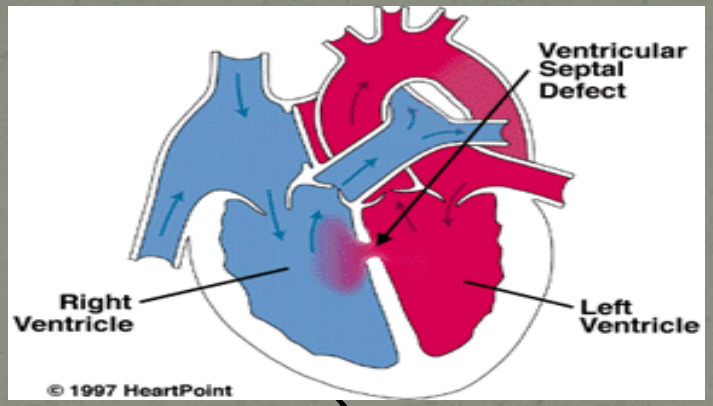
** Transposition of the great arteries.(TGA)

2- **Cyanotic heart disease with decreased pulmonary flow:**

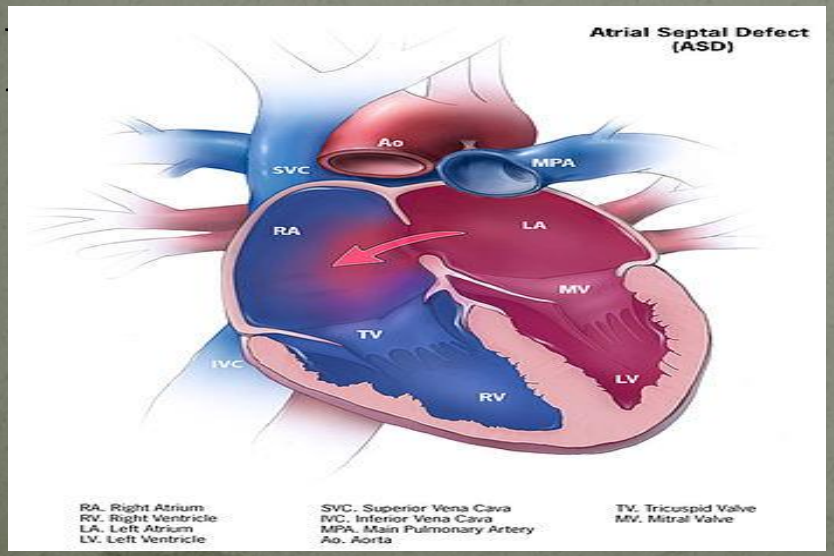
** Tetralogy of Fallot(TOF)

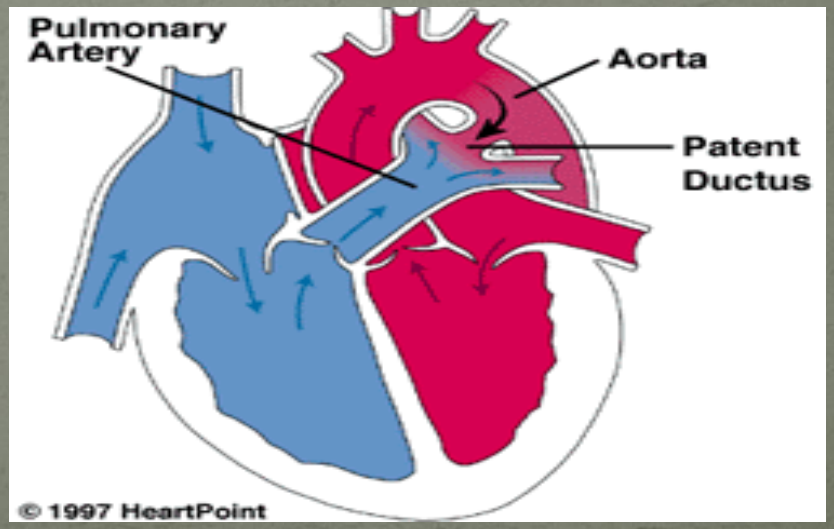
II-Congenital heart disease with potential cyanosis (dominant left to right shunt)

1- Ventricular septal defect (VSD).



2-Atrial septal defects (ASD)-3-Patent ductus arteriosus

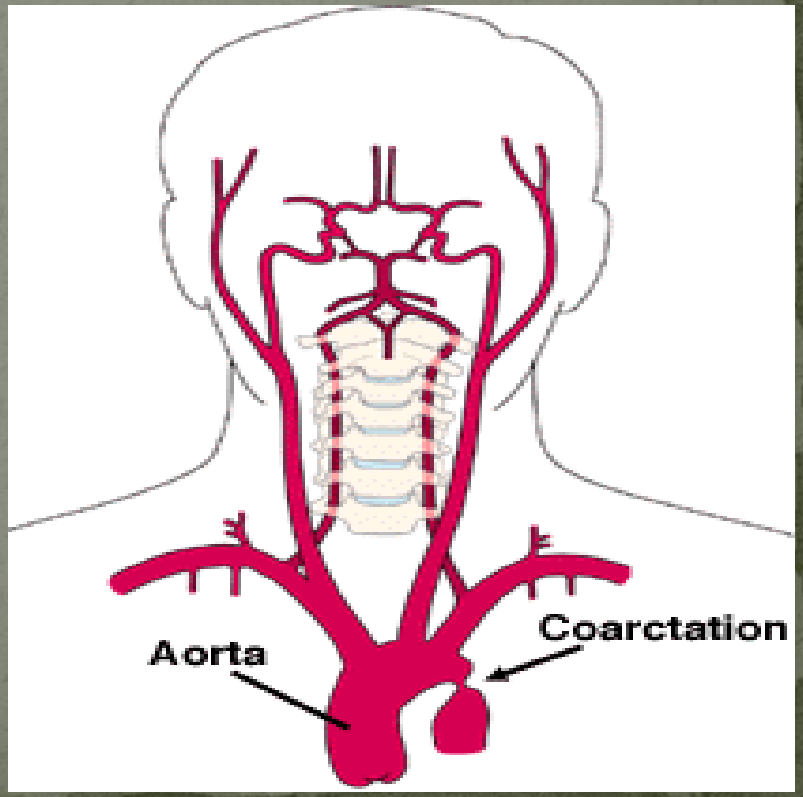




III- Congenital heart disease without cyanosis (i.e. without shunt)

1- Congenital valvular stenosis as Aortic or pulmonary stenosis.

2- Coarctation of aorta.



Cyanotic heart diseases

Clinical picture

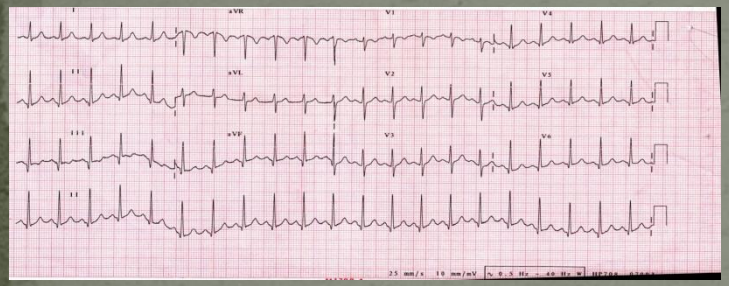
Symptoms:



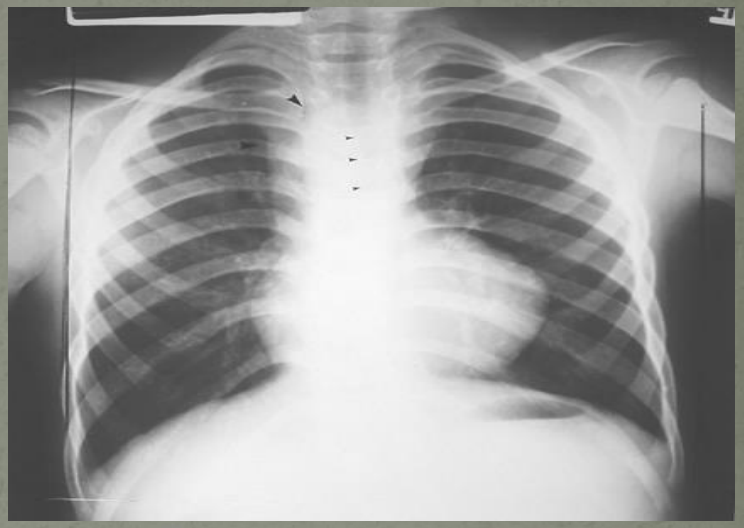
- Cyanosis
- Dyspnea
- Cough
- Paroxysmal cyanotic dyspneic attacks

Investigations

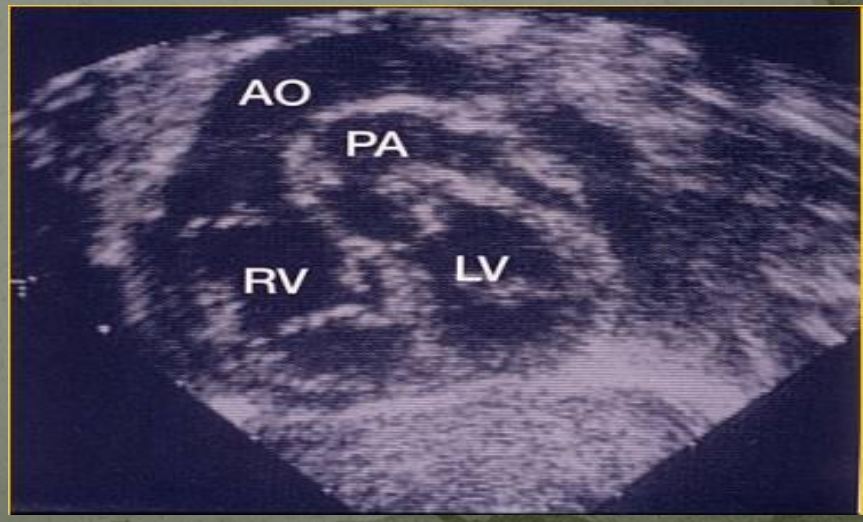
ECG



Radiology:



Echocardiography



-Cardiac Catheterization

Complications:

- Polycythemia
- Cerebral thrombosis & cerebral abscess
- Infective endocarditis

Treatment

1-Medical

a- ttt of hypercyanotic spells

- place in knee-chest position
- O2 inhalation by mask
- IV morphin
- IV sodium bicarbonate
- IV propranolol
- Monitored blood glucose

II-Treatment of heart failure

III-Surgical:

- Surgical repair

Acyanotic heart diseases

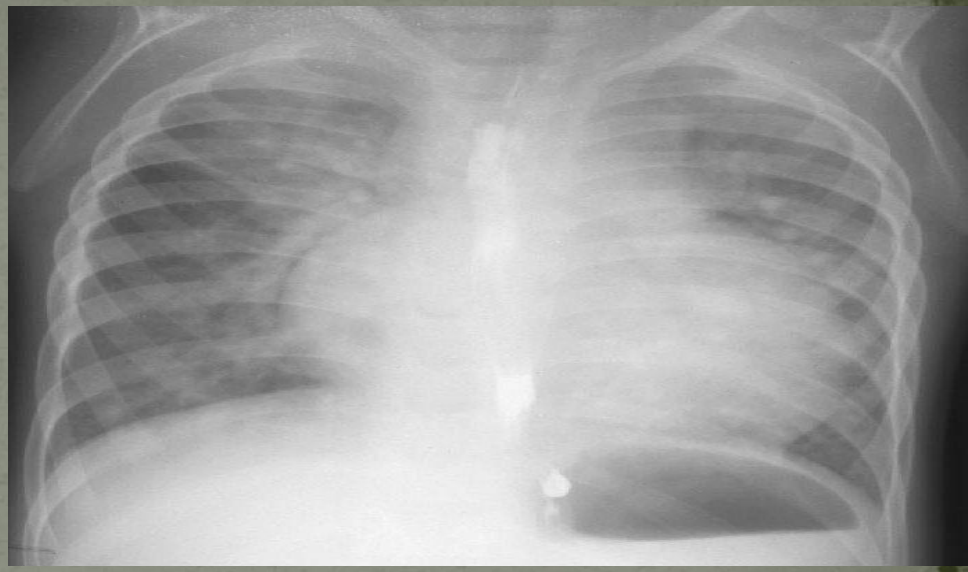
Clinical picture

- Repeated respiratory symptoms
- Episodes of cardiac failure
- Potential cyanosis
- Delayed growth
- Cardiomegaly

Investigations

ECG

Chest X ray



Echocardiography



Treatment

- Medical follow up & ttt of the complication
- Surgical closure for large shunt with recurrent respiratory infection & heart failure

Complications

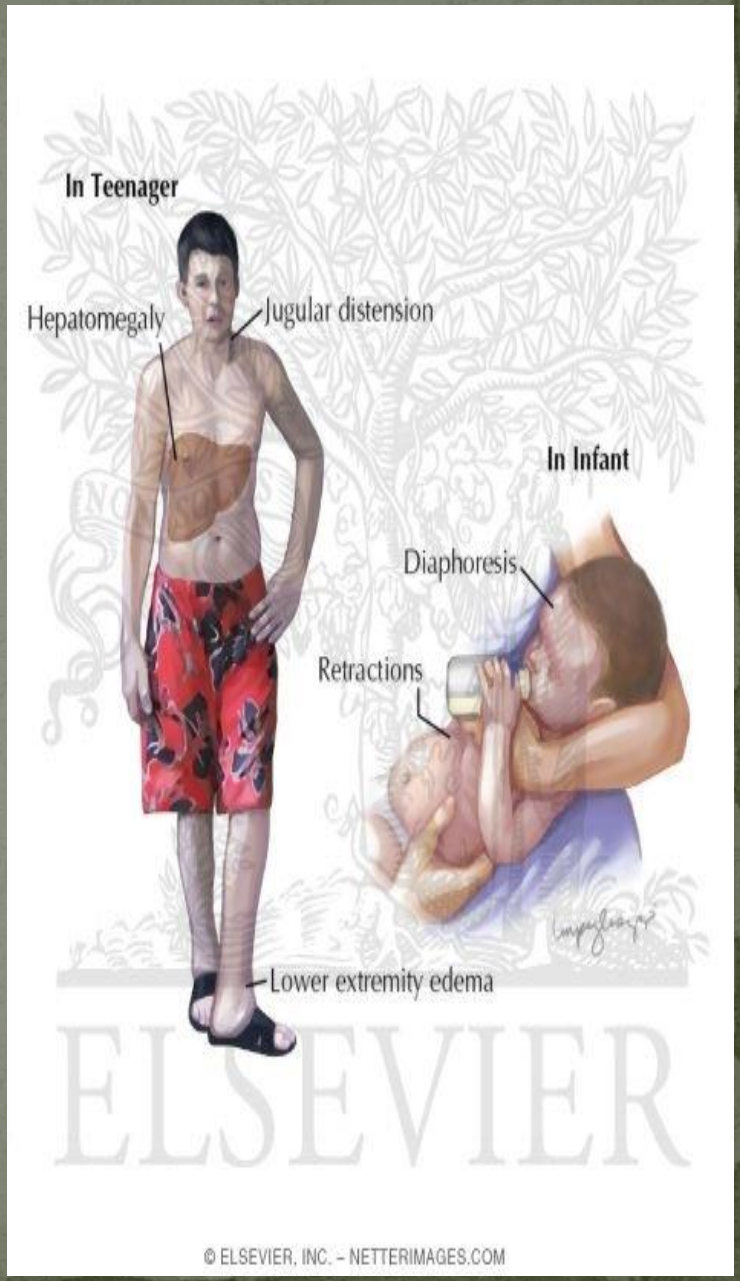
- Infective endocarditis
- Heart failure
- Recurrent respiratory infections
- Pulmonary hypertension, Eisenmenger syndrome

Heart Faliure :

- Heart failure describes a clinical syndrome which results from inability of the heart to adequately deal with the venous return and maintain the cardiac output.

Clinical Picture

- Dyspnea
- Tachycardia.
- Generalized edema and may be ascites.
- Enlarged tender liver.
- Cardiomegaly.



Treatment of Heart Failure



- Bed rest
- Salt Restriction
- Digitalization
- Diuretics
- Inotropic
- After load reducing agents

Digitalization

- **Rapid**

1. Oral :- 0.04mg/kg
2. IV :- 0.03 mg/kg

- ½ immediately
  ¼ 12 hrs later
  ¼ 12 hrs later

- **Maintenance**
  0.01mg/kg/12hrs

- **Slow**
  -maintenance dose
  -post operative
  7-10 days

What to check on Digitalization?

- ECG
- Serum Digoxin level
- Calcium & Potassium levels

(hypokalemia and hypercalcemia ↑ ↑ digitalis toxicity)

Digitalis Toxicity

- **Manifestations**
- **Non Cardiac**

nausea, vomiting,
Visual disturbance,
headache

- **Cardiac**

1.arrythmias
ventricular arryth.
heart block
bradyarrythmias
2.Depressed S-T segment

- **Management**

1. Stop Digoxin
2. Correction of hypokalemia
2. Therapy for ventricular arrhythmias and tachycardia

Diuretics

- **Furosemide**
  - IV 1-2mg/kg/dose /12hrs

    PO 1-4mg/kg/dose 2-three times/day
  - Potassium chloride supplementation
- **Spironolactone** PO 2-3mg/kg/24hrs
- **Chlorothiazide** PO 20-40 mg/kg/24Hrs

After load reducing agents

- **Captopril**

Blocks the production of Angiotension II-

↓ ↓ systemic vascular resistance

Thank you